Justification of disciplinary focus

Team

Group 19: Mazen Hussain, Imran Chowdhury, Fouzan Pasha and Michael Padeigis.

Project: Adaptive Multimodal Wearable for Parkinson’s-Induced Hand Tremor Suppression Using Predictive Control, Mechanical Damping, and Peripheral Electrical Stimulation.

Stakeholder: Parkinson Canada. Co-supervisors: Nicky Anvarihamedani and Mohamad Mirhoseininejad. Instructors: Dr. Vincent Leung and Dr. Spencer Smith.

Motivation and problem identification

Parkinson’s disease is a progressive neurological disorder that can cause resting tremor, rigidity, and impaired movement. Hand and upper-limb tremors can interfere with everyday tasks such as eating, writing, dressing, and typing, reducing independence and quality of life. Existing treatments, including medication and deep brain stimulation, can be limited by variable effectiveness, side effects, cost, or invasiveness. Non-invasive approaches such as peripheral electrical stimulation and wearable mechanical damping offer potential alternatives, but their effectiveness depends on accurately detecting tremor and applying the intervention at the appropriate time. This project therefore aims to develop a patient-adaptive, closed-loop wearable device that senses and characterizes hand tremor caused by Parkinson's in real time and applies a controlled intervention while minimizing interference with voluntary movement.

Solution statement and design approach

The proposed system is a wrist-worn tremor-suppression device integrating motion sensing, signal processing, tremor intervention, and patient-facing software. A six-axis inertial measurement unit will measure upper-limb motion, while signal-processing algorithms distinguish voluntary movement from tremor caused by Parkinson's and estimate characteristics such as frequency, amplitude, and phase. These measurements will support a predictive control system that anticipates upcoming tremor motion and reduces the effect of sensing and actuation delays. Tremor suppression will be investigated using two complementary approaches: semi-active mechanical damping and non-invasive peripheral electrical stimulation through surface electrodes. The mechanical system will apply controllable resistance during tremor while minimizing interference with voluntary movement, while the electrical subsystem will investigate appropriate electrode placement, stimulation timing, and stimulation parameters. Both approaches will operate within the same closed-loop sensing and control framework. A companion mobile application will communicate with the wearable device; the app will monitor status and tremor severity, and store longitudinal tremor data. Users will also be able to record contextual information such as medication timing, sleep, meals, activities, and perceived tremor severity to support longer-term analysis of tremor patterns.

Educational background and expected contributions

The team’s Software and Biomedical Engineering background supports the integration of sensing, computation, and physical intervention into a wearable system for Parkinson’s-induced hand tremors. Fouzan and Mazen will lead data collection and preprocessing, filtering, tremor detection and prediction, embedded firmware, IMU microcontroller integration, and real-time control. They will implement Bluetooth communication and a companion application for recording sensor data, visualizing tremor patterns, and reviewing performance. Testing will assess accuracy, latency, and reliability. Imran and Michael will lead biomedical and physical-system development, including physiological requirements, sensor and actuator

selection, sensor integration, wearable (PCB) design, and stimulation or damping mechanisms. They will develop the tremor-simulation rig and evaluate intervention effectiveness, wearability, and hardware reliability. All members will share responsibility for requirements, safety considerations, prototyping, system integration, testing, and documentation. Previous iBioMed experience with electronics, microcontrollers, motor control, and rapid prototyping supports these contributions. Validation will combine labelled tremor datasets with controlled mechanical experiments, to test and improve the prototype.